



Department of Computer Science and Engineering



Department of
Computer Science and Engineering
National Institute of Technology Calicut

VISION

To be a Centre of Excellence in the Discipline of Computing

MISSION

*To develop human resource with sound knowledge - theory and practical -
in the discipline of computing and the ability to apply the knowledge to the
benefit of the society at large*

ABOUT THE DEPARTMENT

From its inception about a century ago, computing has become the defining technology of our age. The study of computing is the study of the process of abstraction and its power, the practice of design and implementation, mathematical theory of computation and its implications. Undergraduate programme in Computer Science and Engineering started in the year 1985 in the then Electrical Engineering Department. The Department of Computer Science and Engineering (CSE) was established as an independent entity in the year 1997 as a result of trifurcation of the Department of Electrical Engineering. The CSE Department currently offers two MTech programmes and PhD programme in addition to the BTech programme in Computer Science and Engineering. The fast changing scenario in Information Technology necessitates the department to actively extend its research and development activities. The department also strives to fulfil its social responsibility by extending technical support to government and non-governmental organisations through community outreach programmes.

UNDERGRADUATE AND POSTGRADUATE PROGRAMMES

Name of the Degree	Duration	Specialization	Year of Commencement	Total intake
B.Tech.	4 Years	CSE	1985	200
M.Tech	2 Years	CSE	1998	23
M.Tech	2 Years	CSE -IS (Information Security)	2006	22
M.Tech	2 Years	CSE -AIDA (Artificial Intelligence and Data Analytics)	2023	25

Ph.D. PROGRAMME

Year of Commencement	Ongoing	Completed
2002	85	48

PROGRAMME COORDINATORS

B.Tech.	Dr. Vinod Pathari
M.Tech - CS	Dr. Gopakumar G
M.Tech - IS	Dr. P. Arun Raj Kumar
M.Tech - AIDA	Dr. Abdul Nazeer K A
Ph.D.	Dr. Amit Praseed



FACULTY AND SPECIALIZATION

Dr. Abdul Nazeer K A	Ph.D. (NITC)	Artificial Intelligence, Bioinformatics, DBMS
Dr. Amit Praseed	Ph.D. (NITK)	Web Security and Information Security
Dr. Anand Babu N B	Ph.D. (IITB)	Extremal Combinatorics, Graph Theory
Dr. Anil Pinapati	Ph.D. (NITW)	Cryptography and Information Security
Dr. Anu Mary Chacko	Ph.D. (NITC)	Computer Security
Dr. Arun Raj Kumar P	Ph.D. (NITT)	Computer Networks, Internet of Things
Dr. Ashwin Jacob	Ph.D. (IMSc)	Parameterized Algorithms, Kernelization
Dr. B Umamaheswara Sharma	Ph.D. (NITW)	Machine Learning Applications In Software Engineering
Dr. Chandramani Chaudhary	Ph.D. (BITS PILANI)	Machine Learning, Computer Vision and Natural Language Processing
Dr. Gopakumar G	Ph.D. (KERALA Univ.)	Bioinformatics, Data Mining
Dr. Hiran V Nath	Ph.D. (IDRBT-UoH)	Cyber Security and Forensics
Dr. Jay Prakash	Ph.D. (ABV-IIITM)	Data Mining, Soft Computing
Dr. Jayaraj P B	Ph.D. (NITC)	High Performance Computing
Dr. Jimmy Jose	Ph.D. (IITKGP)	Cryptography and Security
Dr. Joe Cheri Ross	Ph.D. (IITB)	Music Information Retrieval, Natural Language Processing
Dr. Lijiya A	Ph.D. (NITC)	Image Processing
Dr. Madhu Kumar S D	Ph.D. (IITB)	Middleware, DBMS, Software Engineering
Dr. Manjusha K	Ph.D. (AMRITA)	Document Analysis, Machine Learning
Dr. Muneeswaran P	Ph.D. (IITMD)	Distributed System, IoT, Formal Concept Analysis
Dr. Murali Krishnan K	Ph.D. (IISc)	Algorithms and Complexity, Coding Theory
Dr. Nirmal Kumar Boran	Ph.D. (IITB)	Computer Architecture, processor Design, Security
Dr. Pournami P N	Ph.D. (NITC)	Image Processing, Machine Learning
Dr. Prabu M	Ph.D. (ANNA Univ.)	Network Security
Dr. Pranesh Das	Ph.D. (NITN)	Data Mining, Machine Learning and Optimisation
Dr. Priya Chandran	Ph.D. (IISc)	Computer Architecture, Algorithms
Dr. Raju Hazari	Ph.D. (IIEST)	Cellular Automata, Machine Learning
Dr. Renjith P	Ph.D. (IIITDM)	Graph Theory & Algorithms, Complexity theory
Dr. S Sheerazuddin	Ph.D. (IMSC)	Theoretical Computer Science (Logic & Verification)
Dr. Saidalavi Kalady	Ph.D. (NITC)	Data Mining, Operating Systems
Dr. Saleena N	Ph.D. (NITC)	Program Analysis, Compilers
Dr. Santosh Kumar Behera	Ph.D. (IITBBS)	Artificial Intelligence, Computer Vision
Dr. Shweta	Ph.D. (IIITA)	Natural Language Processing, Software Engineering
Mr. Sreenu Naik Bhukya	M.Tech. (ANDHRA Univ.)	Network Security, Cryptography
Mr. Srinivasa T M	M.Tech. (VTU)	Bioinformatics
Dr. Subashini R	Ph.D. (IITM)	Algorithms and Data Structures
Dr. Subhasree M	Ph.D. (IITM)	Computational Geometry
Dr. Sudarshan Chakravarthy A	Ph.D. (NITW)	Cloud Computing
Mr. Sumesh T A	M.Tech. (CUSAT)	Computer Networks, Computer Vision
Dr. Vasudevan A R	Ph.D. (NITT)	Computer Security, Network Security
Dr. Veni T	Ph.D. (NITT)	Cloud Computing, Fog Computing
Dr. Venkitarami Reddy	Ph.D. (IITH)	Edge and Cloud Computing, IoT, Machine Learning in Networks
Dr. Vidhya Kamakshi	Ph.D. (IITRPR)	Machine Learning, Deep Learning, Computer Vision
Dr. Vinod Pathari	Ph.D. (IITB)	Information Security Management

SUPPORTING STAFF

Mr. Abdul Sathar V M
Mr. Adhin Suresh S D
Mr. Amit Rag
Mr. Aneesh V
Mr. Arjun K S
Ms. Bhagya Priya B
Mr. Biju Francis
Mr. Brijalal V S
Mr. Mahin Ameer M N
Mr. Rahul N K
Ms. Sreeja V

HEADS OF THE DEPARTMENT SO FAR

Dr. M N Neelakantan	1997 - 2000
Dr. V K Govindan	2000 - 2005
Dr. M P Sebastian	2005 - 2008
Dr. Vineeth Paleri	2008 - 2010
Dr. Priya Chandran	2010 - 2012
Dr. S D Madhu Kumar	2012 - 2014
Dr. K A Abdul Nazeer	2014 - 2016
Dr. Saidalavi Kalady	2016 - 2018
Dr. Saleena N	2018 - 2020
Dr. Muralikrishnan K	2020 - 2022
Dr. Subhasree M	2022 - 2024
Dr. Subashini R	2024 -



LABORATORIES

- Hardware and Embedded Systems Lab
- Secure Computing Lab
- Formal Methods and Algorithms Lab
- Image Processing & Multimedia Lab
- Cloud Computing Lab
- Machine Learning Lab
- Big Data and Biocomputing Lab
- Software Systems Lab
- Network Systems Lab

FACILITIES AVAILABLE

- 350+ Computers – available 24 hours
- High performance computing cluster
- Unique Digital Library support
- Department Eduserver running on Moodle Platform
- Excellent round the clock Internet connectivity
- APJ Abdul Kalam Conference Hall with Video Conferencing facility
- CSED Seminar Hall, Committee Room & Discussion Room
- Class rooms equipped with audio-visual systems
Smart Classroom in NLHC 102

COLLABORATIONS

Research Collaborations

- CMC Vellore, JIPMER Pondicherry
- SCTIMST-Thiruvananthapuram
- Medical Colleges – Kozhikode
- Amala Institute of Medical Sciences
- KMCT Medical College, Malabar Cancer Centre
- Government Medical College Kozhikode
- MVR Cancer Center & Research Institute
- Aster MIMS, JMR InfoTech, Kerala Police

Industry

DELL, Vivid Sparks, NuGenomics, JMR infotech
TCS, SAP, IBM

COMMUNITY SERVICES

Spearheaded the IT literacy programmes at Nadapuram assembly constituency as part of the 'Growing Nadapuram' project

Organized IT awareness programmes for the public at the Koduvally and Chathamangalam Grama Panchayaths

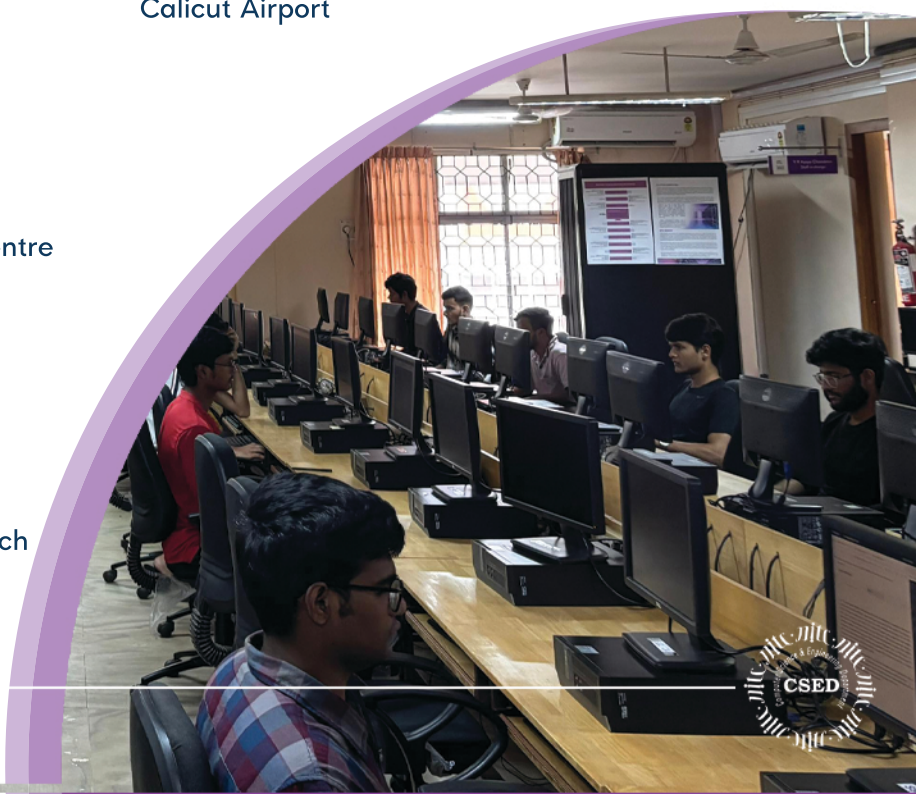
As part of a Community Learning Initiative, 2-day Workshop was conducted on Essence of Computing for high school students

Launched IT@Niravu, a project in collaboration with NIRAVU, a progressive residential forum (NGO) based at Calicut excelling in the field of organic farming

Our students developed an agricultural information web portal named 'Krishippura' to help the organic farmers

Imparted IT training to the rural youth, as part of IT@Niravu

Our students developed an Online Mess Management System for the CISF Unit of Calicut Airport



MAJOR EVENTS ORGANISED

The Computer Science and Engineering Association (CSEA) organised an interactive session about "Career Options After BTech CSE" by Mr. Muhammed Yazar Y, Software Engineer at Facebook, USA on 5 November 2022.

CSEA organised 12th edition of Prof. M.N. Neelakantan Memorial Lecture Series. Mr. Biju Thomas, Manager at Tata Elxsi and alumnus of NIT Calicut talked about the redefinition of broadband experiences and the future of Residential Broadband Gateways on 16th September 2022.

CSEA organised the 11th Lecture in the Prof. M.N. Neelakantan Memorial Lecture Series by Mr. Abins Jose, Senior Solutions Architect in the "Consumer, Media and Technology" vertical of Mindtree on 21st March 2022.

CSEA organised technical talk by Devika S, Software Engineer, Digital experience, Adobe on November 2021.

Dept. of CSE and AI Club, NIT Calicut organised 3-day Workshop on Generative Adversarial Networks (GANArt 2021), from 20-05-2021 to 22-05-2021.

Interactive Webinar on ReactJS concepts and component creation with Shahda Sayed, Microsoft on October 2021

10th Lecture in the Prof. M.N. Neelakantan Memorial Lecture Series organised by CSEA in August 2021 on "Be Relevant. Cloud, Open source, and More - An Industry Perspective", by Mr. Manoj N. Palat, Open Source Lead and Committer, IBM.

CSEA, Institute Innovation Council, and the AI Club conducted "Introduction to Object Detection", a 2-day workshop on Artificial Intelligence from 28th to 29th November, 2021.

AI Club of NITC conducted a 2-day workshop on transformers titled "The Transformer Epiphany" from 18th to 19th December, 2021.

FOSSCell NITC, in association with the Computer Science and Engineering Association, organized Code.init(), the first of its kind DevSprint at NITC in January 2020.

9th Lecture in the MN Neelakantan Memorial Lecture Series conducted on 9th September 2020 with a talk on "Leadership in the 'New Normal'" by Mr. Sunil Plavians, Senior Director - Global Services and Site Head, Finastra.

CSEA activities of 2019-20 inaugurated by Mr. Raveendran Kasthuri, CEO of ULCCS Group, Calicut on 18th September 2019, with a panel discussion on the topic "Industry Institute Collaboration: An Opportunity" and a keynote address on "Tips for takeoff - to a successful IT career" by Mr. Kasthuri.

Department of Computer Science & Engineering jointly with the Department of Radiation Oncology and Medical Physics, MVR Cancer Centre and Research Institute Calicut organised 6-day Faculty Development Programme on "Geometry, Imaging and Parallel Computing Techniques for Cancer Treatment (GIPCAN 2019)", under TEQIP-III from 10th to 15th June 2019.

Smart India Hackathon (SIH-2019), with support from MHRD and AICTE organised in March 2019.

Department of CSE jointly with the Institute of Mathematical Sciences Chennai organised an NCMACM India Summer School on "Graph theory and Graph Algorithms", from 17th June to 5th July 2019.



A FEW DISTINGUISHED ALUMNI

ALUMNI.



Nithin Varma

2011
Faculty member
Max Plank Institute
for Informatics
MPI, Germany



Fahad Panolan

2009
Lecturer
School of Computing
University of Leeds, UK



Jayalal Sarma

2002
Professor
IIT Madras



Sunil Chandran

1996
Professor
IISc Bangalore



Karthik Reghunathan

2008
Director of Research
MindMeld



Preethi Jyothi

2006
Associate
Professor
IIT Bombay



Sajjan Skaria

1996
Oscar winner
Pixar's character supervisor
'Inside Out'



Muhammed Yazar

2014
Software Engineer
Meta, USA



Srijith P.K

2006
Associate Professor,
IIT Hyderabad



Yadu Vasudev

2008
Assistant Professor
IIT Madras



Dipanjan Dutta

1996
Vice president of Engineering
Radisys Cooperation



PC Mustafa

1995
CEO
iD Fresh Foods



Jayafar Moidu

1992
Founder & CEO
JMR InfoTech



Anoop M. Namboodiri

1996
Professor
IIIT Hyderabad



Benny George K

2002
Assistant Professor
IIT Guwahati



Rose Catherine Kanjirathinkal

2007
Research Scientist
Meta AI



Sajith Gopalan

1990
Professor
IIT Guwahati



Ramanujan Sridharan

2009
Associate Professor,
University of Warwick



Maria Francis

2007
Assistant Professor
IIT Hyderabad



Satyadev Nandakumar

2000
Associate Professor
IIT, Kanpur

RESEARCH AT CSED

Outreach activities

1. Three Days National Workshop on SAGE: Software Advancements in the Gen-AI Era
2. Three Days Workshop on Modern Cryptographic Protocols to Resist Quantum Attacks
3. High-End workshop on "Developing a Robust and Secure Routing Algorithm for Real-time Vehicular Networks, January 2024
4. Skill development course on AI for HealthCare, January 2024
5. One Week Advanced Entrepreneurship Skill Development Program on Artificial Intelligence and Machine Learning, March 2023
6. Organized two weeks ATAL-FDP on Foundations of Modern Cryptography and Blockchain, December 2022
7. Five days Faculty Development Programme (FDP) on Continuing Education Programme on Foundations of Modern Cryptography (FoMC-2021) , November 2021
8. NCM-ACM Summer school on Graph theory and Graph Algorithms, July 2019
9. FDP on Geometry, Imaging, and Parallel Computing for Cancer Treatment, June 2019
10. Global Initiative of Academic Networks (GIAN) Course on High Performance Computing, October 2018
11. FDP on Security and Privacy in Big Data Analytics, May 2018

Faculty Research Groups

Algorithms and Formal Methods Group

Social Media Analytics and Cloud Security Group

Machine Learning and Computational Intelligence Group

Bioinformatics, AI and Data Analytics Group

Secure Communication Systems Group

Intelligent Computing Group

Hardware and Security Group

Systems and Verification Group

Computer Vision And Learning Applications

Automated Software Engineering Research Group

PATENTS/ REVENUE GENERATION

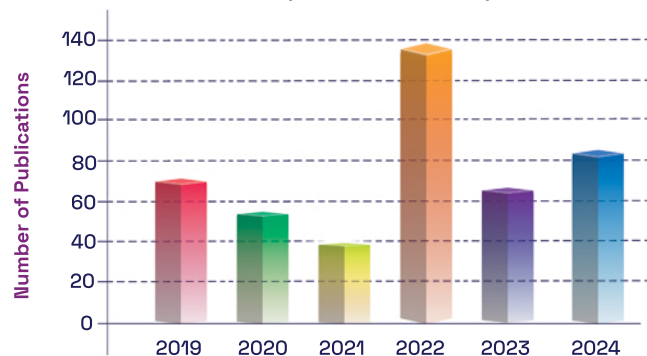
Patents granted: 02

Patents filed: 05

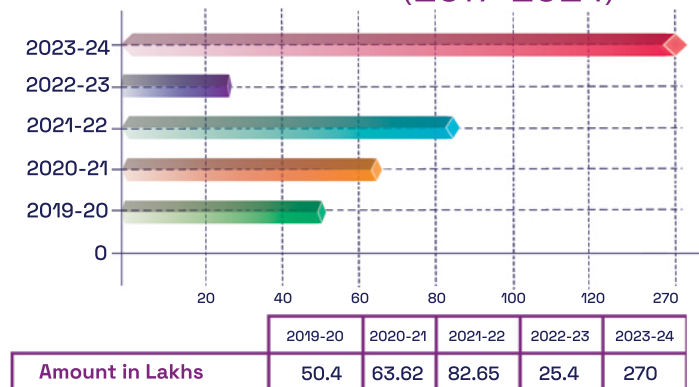
Consultancy Projects: 01

Opportunity: The Department has expertise for consultancy and collaborative research in Cloud computing, Bioinformatics, Cybersecurity, Health Care, AI, and Computer Hardware.

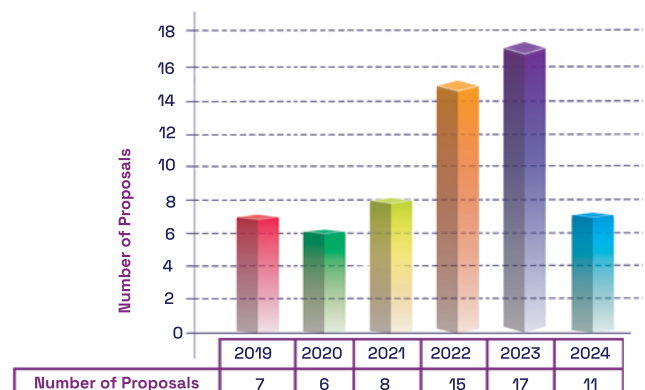
PUBLICATIONS (2019-2024)



FUNDED RESEARCH PROJECTS (2019-2024)



SUBMITTED PROPOSALS (2019-2024)





നാഷണൽ ഇൻസ്റ്റിറ്റ്യൂട്ട് ഓഫ് ടെക്നോളജി കാലികറ്റ്
राष्ट्रीय प्रौद्योगिकी संस्थान कलिकट
NATIONAL INSTITUTE OF TECHNOLOGY CALICUT

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National Institute of Technology Calicut, NIT Campus PO, Calicut, Kerala - 673 601, India
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